

Random Variables, Expectation, and Inequalities

Topic 2

Through this topic, you should...

- Understand The definition of expectation and cumulative density function (CDF).
- Manage probabilistic inequalities and probability bounds.
- Manage finite moment conditions and bounds.

Contents

1. Random variables, laws, and expectation
2. Computing expectations from a CDF
3. Tails, finite moments, and L^p
4. Bounds for transforms, products, and sums
5. Probability bounds from moments

A random variable is a measurable map

Definition 1 (Random variable or random element) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(S, \mathcal{M})$ a measurable space.

Expectation is the Lebesgue integral with respect to $\mathbb{P}$

Once a random variable is measurable, the integral constructed in Week 1 applies on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 2 (Nonnegative expectation) For a measurable $Y : \Omega \rightarrow [0, \infty]$, define

Definition 3 (Extended expectation and integrability) For an extended-real random variable Z , let

A law is a pushforward probability measure

Definition 4 (Law or distribution) Let X be an S -valued random variable.

Expectation is integration against the pushforward law

Theorem 5 (Law of the Unconscious Statistician)

A CDF records the entire law

Definition 6 (Cumulative distribution function)

CDF increments give the Lebesgue integral

Definition 7 (Integration with respect to a CDF) Write $F = F_X$ and let $\mu_F = \mathcal{L}(X)$, uniquely determined by F . Define

Proposition 8 (A formula using only CDF increments) If g is continuous on $[a, b]$ and partitions $a = x_0 < \cdots < x_m = b$ have mesh $\max_i (x_i - x_{i-1}) \rightarrow 0$, then

Integration by parts computes from the CDF itself

Theorem 9 (CDF integration by parts) Let $a < b$ and $g \in C^1([a, b])$, extended by zero outside $[a, b]$ when evaluating $g(X)$. Then

Expectation of a truncated random variable

Corollary 10 For any real random variable X and finite $a < b$, define the truncated

Positive and negative CDF tails determine the mean

Theorem 11 (Expectation directly from a CDF) For every real-valued random variable X ,

A mixed CDF contributes both a jump and an interval integral

Example 12 (Calculate from the CDF without assuming a density) Suppose the input is

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{2}, & 0 \leq x < 1, \\ x/2, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

Inverse transform constructs every real law

Definition 13 (Generalized quantile) Let F be nondecreasing, right-continuous, with limits 0 and 1. For $u \in (0, 1)$ define

$$Q_F(u) := F^{\leftarrow}(u) := \inf\{x \in \mathbb{R} : F(x) \geq u\}.$$

For a random variable X , write $Q_X := Q_{F_X}$.

Inverse-transform method If $U \sim \text{Unif}(0, 1)$ and $X = Q_F(U)$, then $F_X = F$. The key equivalence is

$$Q_F(u) \leq x \iff u \leq F(x).$$

A uniform variable generates an exponential law

Example 14 (Inverse transform and expectation) For $\lambda > 0$, consider $X \sim \exp(\lambda)$,

A quantile can reveal exactly which moments exist

Example 15 (A Pareto variable) On $(\Omega, \mathcal{F}, \mathbb{P}) = ((0, 1), \mathcal{B}((0, 1)), \text{Leb})$, set $U(\omega) = \omega$ and, for $\alpha > 0$,

$$X_\alpha = (1 - U)^{-1/\alpha}.$$

Absolute continuity is exactly the condition behind a density

Definition 16 (Absolutely continuous function) A function $F : \mathbb{R} \rightarrow \mathbb{R}$ is absolutely continuous if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that every finite pairwise disjoint collection of intervals (a_k, b_k) satisfying

$$\sum_k (b_k - a_k) < \delta$$

also satisfies

$$\sum_k |F(b_k) - F(a_k)| < \varepsilon.$$

Lebesgue FTC used below Such an F has an a.e. derivative satisfying Conversely, an indefinite

integral of an L^1 function is absolutely continuous.

Theorem 17 (CDF-density equivalence) A real random variable X has a Lebesgue density if and only if F_X is absolutely continuous. In that case $f_X = F'_X$ almost everywhere.

The Cantor set and its binary encoding

Definition 18 (Middle-thirds Cantor set) Start with $C_0 = [0, 1]$. To obtain C_{n+1} , delete the *open middle third* of every interval in C_n , keeping all endpoints. Thus

$$C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1],$$

$$C_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1].$$

Define $C := \bigcap_{n \geq 0} C_n$. Each C_n contains 2^n disjoint closed intervals of length 3^{-n} . Hence C is closed, and

$$0 \leq \text{Leb}(C) \leq \text{Leb}(C_n) = (2/3)^n \rightarrow 0.$$

A continuous CDF need not have a density

Counterexample 19 (Cantor distribution)

Three kinds of mass in a real probability law

Theorem 20 (Atomic, singular continuous, and absolutely continuous parts) Every probability measure μ on $\mathbb{R}$ decomposes as

Every positive moment is an integrated tail

Corollary 21 (Moment-tail identity)

Definition 22 (L^p and L^∞)

Every finite convex function has a supporting line

Lemma 23 (Supporting lemma) Let $I \subseteq \mathbb{R}$ be an open interval and let $\varphi : I \rightarrow \mathbb{R}$ be convex. For every $m \in I$, there exists $a \in \mathbb{R}$ such that

$$\varphi(x) \geq \varphi(m) + a(x - m), \quad x \in I.$$

Jensen follows by integrating the supporting line

Theorem 24 (Jensen's inequality) Let $\varphi : I \rightarrow \mathbb{R}$ be convex on an open interval.

Monotonicity of L^p space

Hölder is the normalized form of scalar Young

Theorem 25 (Hölder's inequality)

Minkowski for $p > 1$ is one Hölder calculation

Theorem 26 (Minkowski's inequality)

A moment estimate can control an event

The preceding inequalities bound expectations of transforms, products, and sums. How to bound $\mathbb{P}(|W| \geq t)$?

Generalized Markov inequality

Proposition 27 (Generalized Markov inequality) Let $X \geq 0$, let $\psi : [0, \infty) \rightarrow [0, \infty)$ be nondecreasing, and let $t \geq 0$ satisfy $\psi(t) > 0$. Then

Proposition 28 (Tail decay for one integrable moment) If $X \in L^p$ for some $p > 0$, then

Faster tail decay alone need not give a finite moment

Counterexample 29 (The logarithmic boundary case) Fix $p > 0$. Define a law by its tail

$$\mathbb{P}(V_p > t) = \begin{cases} 1, & t < e, \\ \frac{e^p}{t^p \log t}, & t \geq e. \end{cases}$$

Theorem 30 (Cantelli's inequality)